

ICT2217 Network Security  
AY2025/26 Trimester 3  
Team Project



Weightage: **35%**

**Team Formation:** 5 ± 1 students of your choice. Please go to LMS ICT2217 course site to self-enroll your team by **Tuesday 19 May 6:00 pm**.

Due Dates:

**Report + Program:** **Sunday 28 Jun 23:59**, submission via **LMS Dropbox** according to your team number.

**Demo:** **Tuesday 30 Jun** during lab class.

**PART I: Investigating and Demonstrating Real-World Network Attacks and Defenses**

**A. INTRODUCTION**

You have started experimenting various network attacks and defenses in ICT2217 lab exercises. Hope that you have enjoyed, and most importantly have a better understanding of how networks can be attacked and defended in the real world.

Of course, there are more network attacks and defenses that we do not have time to cover them all in the lab exercises. So with the foundations that you've learnt, now is your opportunity to **investigate other real-world network attacks and defenses** of your interest, and to **demonstrate your knowledge and skills** in conducting in the lab environment.

**B. LEARNING OUTCOMES**

Upon completion of this assignment, you should be able to:

- Analyse and demonstrate real-world attacks over computer networks and configure suitable defenses on network devices to mitigate the attacks.

**C. REQUIREMENTS**

Your attacks and defenses must be done over a network. For example, your team may:

- Combine/Extend network attacks and defenses that you've learnt in the lab exercises; or
- Investigate real world network attacks and defenses that have caught your interest but was not covered; or
- Design and experiment with your own network attacks and defenses

As the saying goes – the sky's the limit! Or does the sky have a limit? Enjoy!

## **PART II: Programming Security Tool using Python with Scapy**

### **A. INTRODUCTION**

You have been using the security tools in Kali Linux for your lab exercises. However, to be IT security professional, we cannot behave like script kiddies who just wait for someone to program the tools that we need. Hence in this Part II, you are going to try programming a security tool yourself!

Programming security tools requires the ability to manipulate the values in the protocol headers to send out malicious frames/packets to attack. Nowadays, manipulating the values in the protocol headers is made simpler with the availability of add-on packages/libraries, e.g. Scapy in Python.

For an introduction to Scapy, please read:

Download Scapy: <https://scapy.net/>

Scapy's documentation: <https://scapy.readthedocs.io/en/latest/>

Build your own tools with Scapy:

<https://scapy.readthedocs.io/en/latest/extending.html>

### **B. LEARNING OUTCOMES**

Upon completion of this assignment, you should be able to:

- Program a security tool using Python with Scapy based on theoretical understanding of network operations and attacks covered in ICT1013 and ICT2217

### **C. REQUIREMENTS**

Depending on the design of your attacks in Part I, you may be using one or more security tools in Kali. Instead of using all the existing tools in Kali, your program must:

- Be used to conduct **at least ONE of the sequences of attacks** in Part I

**Note:** No mark will be awarded for GUI. So your program may simply be run using command line since this is a module on Network Security, and not HCI.

**SUBMISSION AND GRADING FOR PART I and II:**

Report covering:

- Theoretical explanation of your network attacks and defenses
- Detailed instructions on your lab network topology setup and procedure to demonstrate the attacks and defenses
- How your program is run to perform the attack(s)
- Relevant Wireshark captures to show your program is indeed sending out the malicious attack frames/packets as intended
- Relevant screen shots as evidences to show your network attacks and defenses are successful

Demo covering:

- Practical setup of your lab network environment and successful demonstration of your network attacks and defenses

Text files containing:

- Source codes of your program

Grading scheme:

- How realistic and creative are your network attacks and defenses
- Theoretical understanding of how the network attacks and defenses work
- Practical skills in demonstrating the network attacks and defenses
- Design logic, programming style, and readability of your program

**Note:** Your program should not be found from the Internet (i.e. NO plagiarism, do NOT simply copy and amend programs found from the Internet and submitted as yours.) In fact programs that are more complicated than necessary will get lower grades. So use this assignment as a learning opportunity to practice and write the tool yourself.